



IDBI INNOVATE

2026

Build. Integrate. Transform.

Knowledge Partner



Powered by



Technology Partner



Team Details

- a. **Team name:** Black Bulls
- b. **Team leader name:** Shubham
- c. **Problem Statement:** Open Track — AI copilot for channel-sourced retail lending (conversational AI · product advisory · financial-health scoring · default prediction)

Brief about the idea

A channel-aware loan-origination & partner-relationship copilot — on top of IDBI's LOS / BRE / LMS

Retail lending is sourced through DSAs & dealers and runs on TAT + trust. We work all four sides at once:



Customer

Only suitable, clearly-explained, instant offers — no spam, no black box.



RM / Bank

A queue ranked by risk-adjusted profit, auto-scoring & recommendations.



Channel partner (DSA)

Instant decisions, transparent payouts, tiering — the loyalty moat.



Risk / Regulator

Explainable, fair, adverse-selection-aware lending with audit trail.

One line: Lend faster, safer, and win partner loyalty.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it's different

- ◆ Channel-first, not customer-only — models the DSA relationship: adverse-selection, TAT, tiering, churn.
- ₹ Bank-grade economics — ranks by $P(\text{convert}) \times (\text{net interest income} - \text{expected credit loss})$.
- ▲ Explainable — per-decision reason codes + Reg-B / RBI adverse-action codes; fair-lending exclusions.
- Integrates, not replaces — an intelligence layer on top of LOS / BRE / LMS.

How it solves the problem

- ✓ Right lead → right product → right price, instantly, with a professional offer.
- ✓ Flags risky sourcing before disbursal → fewer NPAs.
- ✓ Transparent, fast, tiered payouts → partner loyalty → more & better volume.
- ★ The flywheel: win the partner → volume → data → sharper models.

List of features offered by the solution

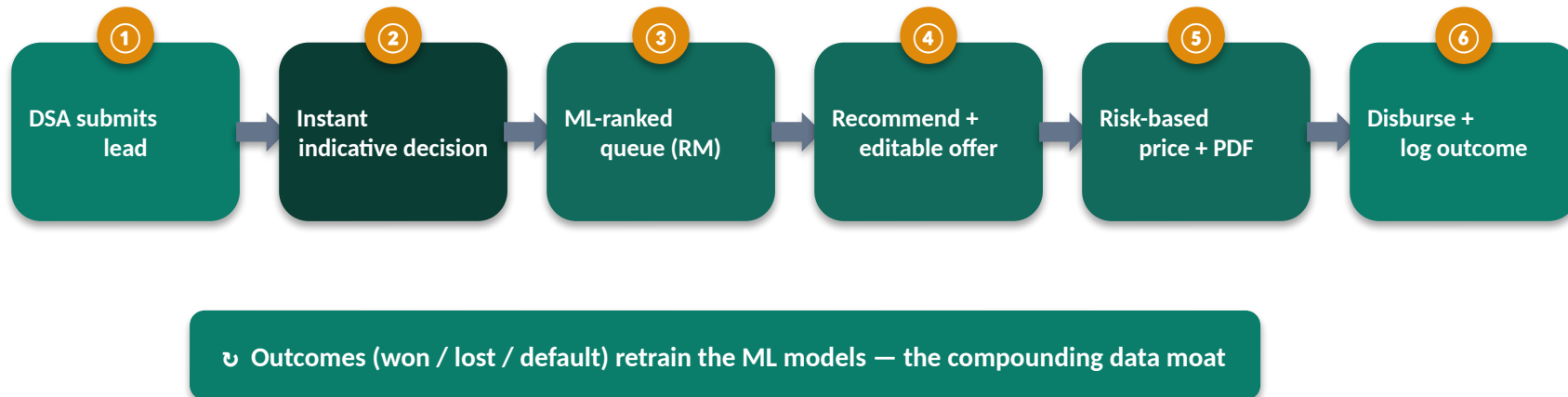
Customer & RM

- ✓ ML conversion-propensity queue (risk-adjusted)
- ✓ Explainable financial-health score (0-100)
- ✓ Deterministic eligibility, rate & EMI engine
- ✓ Suitability-first next-best-product advisory
- ✓ Editable offer + risk-based pricing in-band
- ✓ Premium branded PDF + WhatsApp / Gmail
- ✓ Per-decision reason codes on every score

Channel & Partner

- ✓ Partner/lead default-risk (adverse-selection)
- ✓ Channel-mix ROI + book concentration (HHI)
- ✓ Commission + clawback payout engine
- ✓ Duplicate / fraud lead detection
- ✓ Instant decision (<50 ms) + SLA / TAT clock
- ✓ Partner tiering + churn early-warning
- ✓ Agentic AI assistant (natural language)

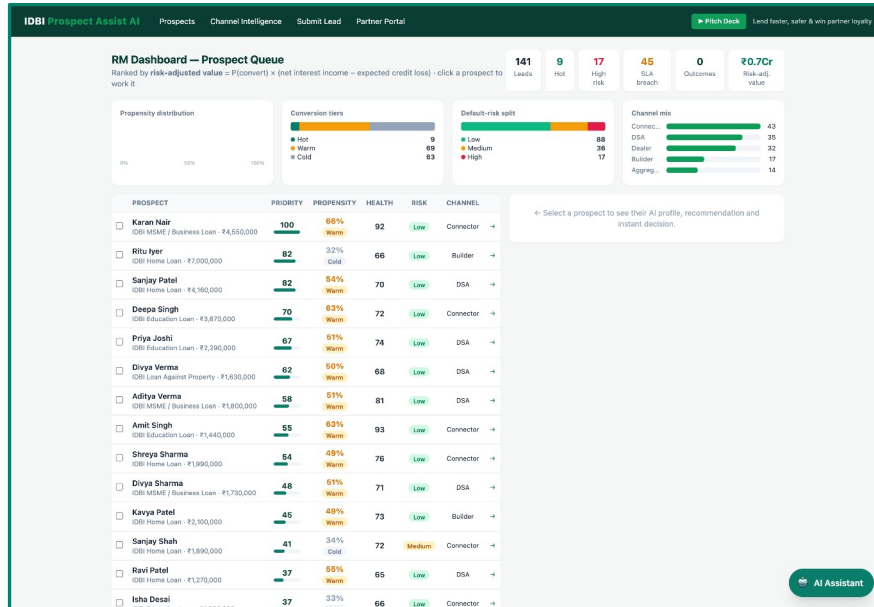
Process flow diagram or Use-case diagram



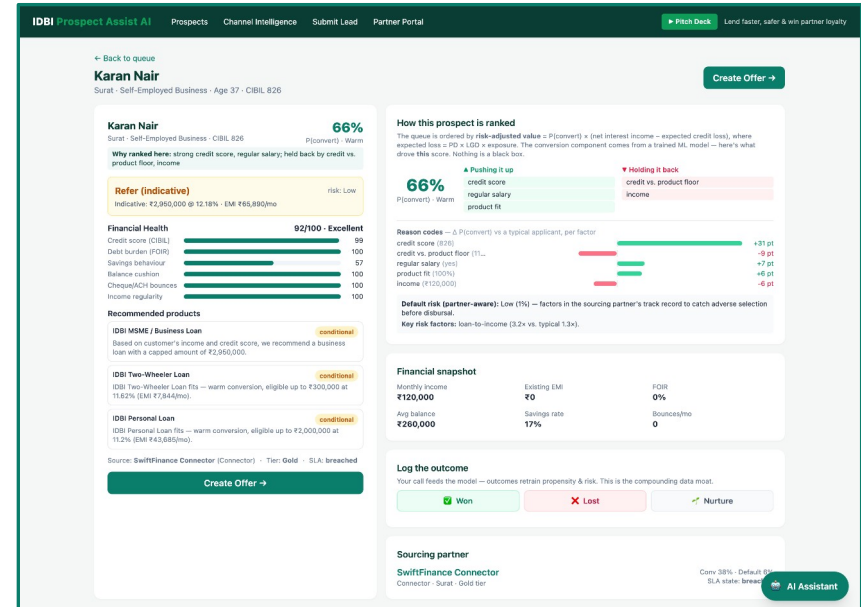
Numbers boundary: the deterministic engine owns every rupee (eligibility, rate, EMI, ECL); ML owns probabilities; the LLM only phrases. RM stays in control — edits re-validate through the engine.

Wireframes/Mock diagrams of the proposed solution (optional)

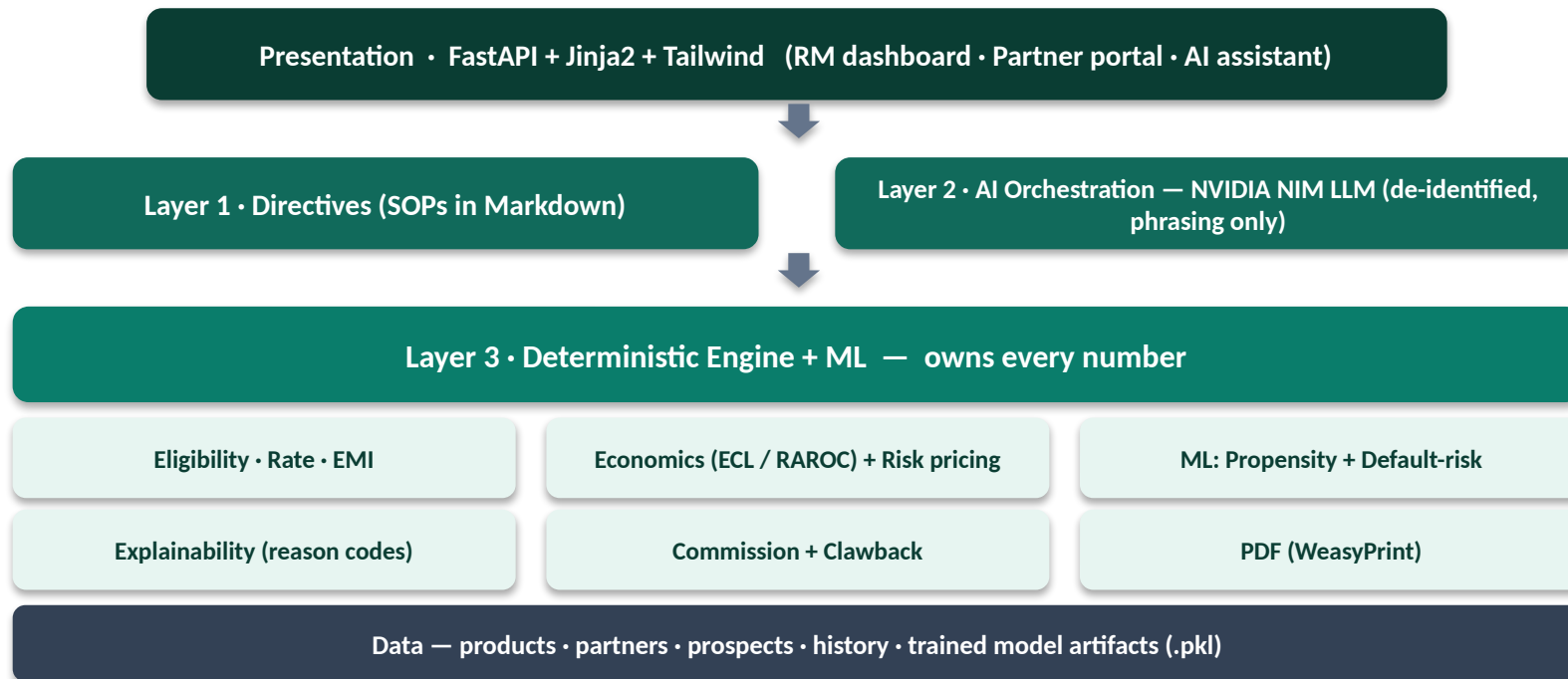
RM dashboard — risk-adjusted, ML-ranked queue



Prospect profile — reason codes & recommendation



Architecture diagram of the proposed solution



Integration points (PoC): CIBIL bureau • KYC / e-consent (DPDP) • LOS / BRE / LMS • NVIDIA AI Enterprise / on-prem

Technologies to be used in the solution

Backend & Web

Python 3.14

FastAPI

Uvicorn

Jinja2

Tailwind CSS

vanilla JS

Machine learning & AI

scikit-learn

NumPy / SciPy

joblib

occlusion XAI

NVIDIA NIM LLM

Documents & PDF

WeasyPrint

pango / cairo

Quality & Deployment

pytest · 44 tests

Playwright

Docker

Render / Railway / Cloud Run

Estimated implementation cost (optional)

Prototype — today

≈ ₹0

Open-source stack
NVIDIA NIM free tier
Free-tier hosting • no licence cost

Production PoC — indicative (annual)

- ▶ LLM / GPU inference — usage-based
- ▶ Cloud hosting + DB + storage — modest
- ▶ Integration — LOS/BRE/KYC/CIBIL (one-time)
- ▶ Model-ops — monitoring, retraining, governance

ROI: cost saved (NPA reduction + payout-leakage → 0 + RM selling hours) >> cost to run

Snapshots of the prototype

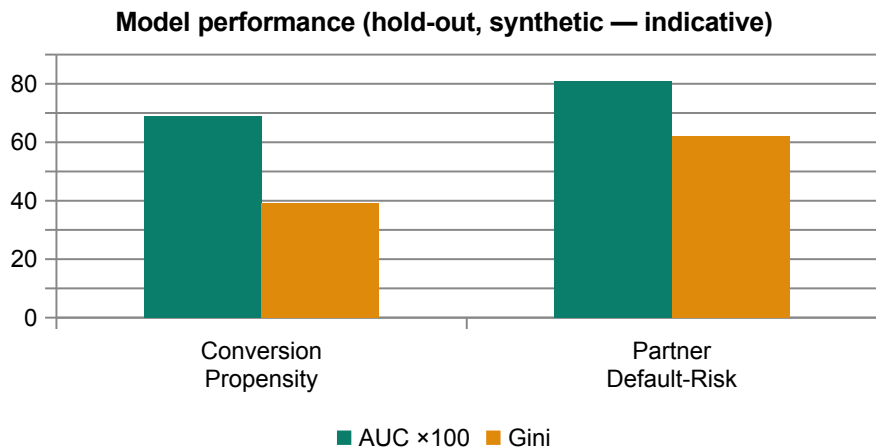
Editable offer — live EMI & risk-based pricing

Instant decision + adverse-action reason codes

Channel intelligence — adverse selection, ROI, HHI

Built-in pitch deck (live numbers)

Prototype Performance report/Benchmarking



< 50 ms

Instant decision latency

~1-2 s

LLM reply (9 s timeout → fallback)

0.22

Partner-history feature share — no target leakage · Bayesian shrinkage

Business impact on the synthetic book

₹16.4 Cr
pipeline value

₹0.68 Cr
risk-adjusted value

44 / 44
automated tests passing

Additional Details/Future Development (if any)

Phase 1 • Integrate

- ▶ IDBI sandbox APIs
- ▶ CIBIL bureau + KYC
- ▶ e-consent (DPDP Act)
- ▶ LOS / BRE / LMS hooks

Phase 2 • Harden

- ▶ Retrain on real conversion / NPA / TAT
- ▶ SHAP + PSI drift monitoring
- ▶ Auth / RBAC + audit trail
- ▶ Data residency

Phase 3 • Scale

- ▶ Partner mobile portal + WhatsApp bot
- ▶ OTP-gated secure delivery
- ▶ Collections & Early-Warning (EWS)
- ▶ NVIDIA AI Enterprise / on-prem

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link



GitHub Public Repository

<https://github.com/Shubham-33/Hack2Skill-IDBI-Hackathone>



Final Product Link (live prototype)

<https://prospect-assist-ai-2guf.onrender.com>



Demo Video Link (3 minutes)

[paste your unlisted YouTube / Drive link]



IDBI 2026 INNOVATE

Build. Integrate. Transform.

Knowledge Partner



Powered by



Technology Partner



Thank you!